



A numerical study on the buoyancy effect around slanted-pin fins mounted on a vertical plate (Part-II: Laminar mixed convection)

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ABSTRACT

This paper presents a numerical study on the mixed convection heat transfer of slanted-pin fins mounted on a vertical heated wall for the laminar flow regime, which is an extension of the authors' accompanying study (Oh et al. 2018) on the natural convection. Using the ideal gas assumption for the air, three-dimensional governing equations for fluid flow and heat transfer were calculated by imposing a periodic boundary condition in the horizontal lateral direction with the SIMPLE algorithm. The discrete ordinate method was also included to calculate the effect of radiation. The effects of the fin-inclination angle were investigated in a modified Richardson number range of 500–2500 by varying the Reynolds number at a fixed Grashof number. It was found that as the forced convection increases, the fin-inclination angle showing the best heat transfer performance changes from negative to positive fin angles. The critical Richardson number for this transition and the corresponding flow and heat-transfer characteristics in the mixed convection regime are summarized.

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1. Introduction

Improving the heat transfer performance is a subject of considerable interest in the design of thermal systems. Heat sinks with extended surfaces, such as plate- or pin-shaped fins, have been used to enhance the heat transfer by increasing the surface area. The heat transfer performance of these heat sinks can be augmented by buoyancy-driven secondary flows [1–6] and the forced flows [7–12]. In the mixed convection regime, however, the heat sink shows complex flow characteristics depending on the relative influence of natural- and forced-convection effects. There have been various studies of the cooling performance capabilities of heat sinks under mixed-convection conditions.

Kobus and Oshio experimentally investigated the thermal performance of a staggered vertical pin-fin array heat sink under mixed convection with an impinging flow [13] and an assisting flow [14]. They studied the influence of thermal radiation on the cooling performance of a pin-fin heat sink and investigated the effects of several geometric parameters, such as the fin diameter, height, and spacing, on the effective thermal resistance. The effects of variable properties under the natural convection conditions from a vertical plate were experimentally investigated by Sieber

et al. [15]. They found that the transition from a laminar to a turbulent flow for the heated wall temperature condition is closely related to the Grashof number and the temperature ratio. Abedin et al. [16,17] carried out a direct numerical simulation for mixed convection heat transfer with flows aiding and opposing the gravity. Chen et al. [18] experimentally and numerically investigated the mixed convection heat transfer and fluid flow characteristics of a plate-fin heat sink in a wind tunnel. They found secondary vortices in specific conditions and regions. In addition, the effects of the fin spacing and the magnitude of the inlet velocity on the pressure drop were examined at selected measurement locations. Jeng [19] experimentally studied the mixed convection heat transfer of a novel finned heat sink in vertical flow passages of a LED lamp. The effects of the fan motor and the vertical passages as well as the degree of separation completeness were analyzed and an empirical formula for the heat transfer performance was proposed. Deshmukh and Warkhedkar [20] experimentally investigated the thermal performance of an elliptical pin fin heat sink under mixed convection with an assisting flow. They carried out comprehensive experimental investigations of the air-side thermal performance of fully shrouded, elliptical and circular pin fin heat sinks under mixed convection. Dogan and Sivrioglu [21] investigated mixed convection heat transfer from longitudinal fins in a horizontal rectangular channel experimentally. They showed that the optimum fin spacing depends on the Rayleigh number for a fixed channel

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Nomenclature

Symbols

A	surface area
AR	aspect ratio
C_p	specific heat
g	gravity acceleration
Gr	Grashof number
H	fin height
h	heat transfer coefficient
k	thermal conductivity of air
L	heat sink length
L_f	fin side length
M	molecular weight
P	fin pitch
p	static pressure
Pr	Prandtl number
q''	heat flux
Q	heat transfer rate
R	universal gas constant
Ra	average Rayleigh number
Ra^*	modified Rayleigh number
S	fin spacing
t	fin thickness
T	static temperature
T^*	dimensionless temperature ratio
u_i	dimensionless velocity components in i -direction
U	Inlet velocity

w	periodic repeat length
x_i	Cartesian coordinate system, $x_i = (x, y, z)$

Greek symbols

α	thermal diffusivity
β	thermal expansion coefficient of air
ε	emissivity
μ	fluid viscosity
θ	fin installation angle
ν	fluid kinematic viscosity
ρ	fluid density

Subscript

<i>base</i>	baseline (plate-fin) results at $\theta = 0^\circ$
<i>cond</i>	conduction
<i>in</i>	velocity inlet
<i>rad</i>	radiation
<i>surw</i>	surrounding air wall
<i>t</i>	total
<i>w</i>	heated wall
<i>i, j</i>	tensor notation
∞	ambient

Superscript

*	normalized value
–	averaged value

height and constant mass flow rate. Mokhtari et al. [22] carried out a study of the mixed convection heat transfer of various fin arrangements in a horizontal channel. They used three types of the heat sinks made from a combination of inclined fins in the span-wise direction and fin arrangements. Depending on the arrangement, significantly improved cooling performance of the base plate was observed due to the flow structures of the air stream under the influence of the fin angle. Yang et al. [23] studied mixed convective cooling performance outcomes to analyze the influence of the orientation of the channel. In a parallel-plate channel, the orientation effect becomes weak as the forced convection increases.

In the authors' accompanying paper [27], a numerical study on the buoyancy effect around slanted-pin fins mounted on a vertical heated plate was conducted for a natural convection condition. For positively inclined fins, an accelerated flow into the fin-stem region, which enhances the heat transfer performance on the base plate, was observed, similar to the results with forced convection cases by Kim et al. [7,8]. For negatively inclined fins, however, the penetration of cold air from the quiescent region into the vertical fin interspacing was observed. This flow enhances the distribution of the heat transfer coefficient on the surface in the vertical fin interspacing. As a result, negatively inclined fins showed the best heat transfer performance, which is opposite to the forced convection result.

The present paper aims to extend the authors' previous study to the case of the laminar mixed convection regime. Because the results of the best heat-transfer-performance fin-inclination angle were opposite to each other for the forced- and natural-convection conditions, the transition phenomena need to be analyzed and the factors critically affecting the variation should be evaluated. The Richardson number, which represents the importance of natural convection relative to forced convection, was varied to determine the effect of the fin-inclination angle on the heat transfer performance. The flow and heat transfer characteristics, such as the

pressure, the friction and the heat transfer coefficients, were quantitatively evaluated to predict the cooling performance of a heat sink with slanted-pin fins.

2. Numerical method

2.1. Computational domain and geometrical configurations

The straight and narrow domain shown in Fig. 1(a) is the simplified computational domain used to investigate the heat transfer performance of a heat sink with fins having various inclination angles under given conditions. Periodic boundary conditions were imposed on the lateral side of the computational domain. Fig. 1(b) shows the boundary conditions for the laminar mixed convection problem considered in this paper. This domain is consistent with the authors' previous study [27] on the natural convection except for the uniform upward inflow U_{in} for finite length L_{in} at the bottom boundary. A symmetry condition was applied to the boundary parallel to the heated plate, as in the previous work [27]. For the solid surfaces of the plate and fins, no-slip conditions were applied. The inside of the fin, which is a solid region, was modeled to calculate the conduction effect of the conjugate heat transfer problem. An ambient boundary condition was imposed on the other boundaries.

The laminar flow condition was referenced from study of Sieber et al. [15]. In this paper, the maximum Grashof number and temperature ratio are correspondingly $Gr = 4.8 \times 10^7$ and $T_w/T_\infty = 1.14$, which are lower than the critical values of the transition in a mixed convection condition. In addition, the condition of the model includes an aiding flow, resulting in greater delay in the transition from the laminar to the turbulent flow under mixed convection according to Abedin et al. [16,17].

Fig. 2 shows the geometric configuration of the fins considered in this study. The horizontal fin with the zero-inclination angle shown in Fig. 2(a) is the conventional pin-fin heat sink model that was used as a baseline for the heat transfer performance

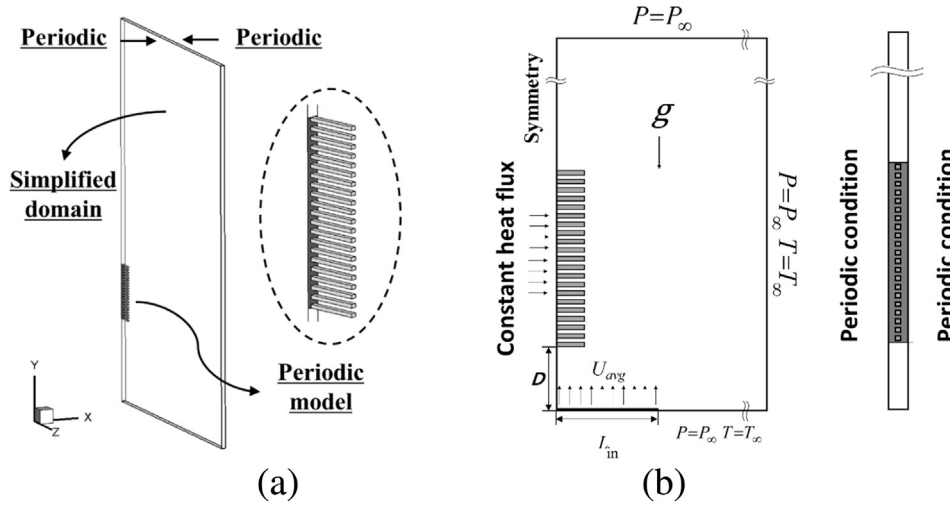


Fig. 1. Computational domain and boundary conditions in this study: (a) simplified computational domain and (b) boundary conditions under external mixed convection. (Not to scale).

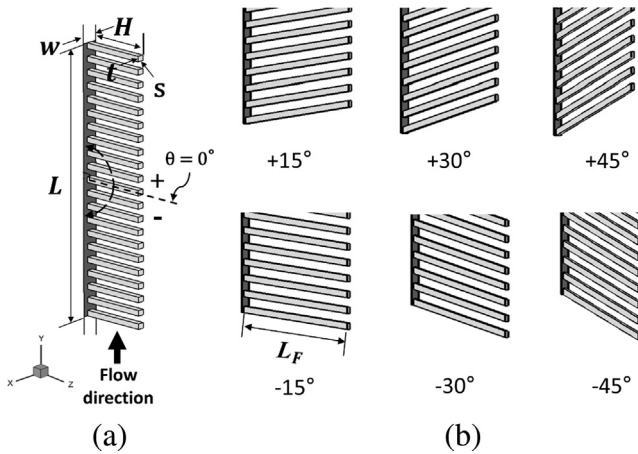


Fig. 2. Geometric configuration of the fins considered in this study: (a) horizontal fins, and (b) inclined fins.

comparison. Fig. 2(b) shows the inclined-fin models tested in the study. The positively inclined fins were slanted in the streamwise direction of the flow and the negatively inclined fins were angled toward the upstream direction. The tested range of the fin-inclination angle is -45° to $+45^\circ$. The fin thickness t and pitch p are 4 mm and 6 mm, respectively. The aspect ratio of the pin-fin $AR = t/s$ is unity, as shown in Fig. 2(a). The length of plate L is 200 mm and the width of plate w is 10 mm, which is the lateral fin pitch. To examine the fin-inclination effect without altering the total heat transfer area, the height of the fin H at a given inclination angle θ was defined as follows:

$$H = \frac{A_{pin-fin, \theta=0^\circ}}{(s + t/\sin \theta)} \quad (1)$$

Table 1 summarizes the calculation conditions for the models considered in this study.

2.2. Governing equations

Assuming a steady state, laminar flow, and constant properties except for the fluid density, the flow and the temperature field around the heated plate with fins are governed by the conservation equations of mass, momentum and energy:

Table 1

Test conditions considered in this study.

Conditions	Square pin-fin models
θ	$-45^\circ, -30^\circ, -15^\circ, 0^\circ, +15^\circ, +30^\circ, +45^\circ$
AR	1
Gr^*	1.47×10^{10}
Re	0, 2422, 2708, 3127, 3830, 5416
$Ri^* = Gr^*/Re^2$	$\infty, 2500, 2000, 1500, 1000, 500$

$$\frac{\partial}{\partial x_i}(\rho u_i) = 0 \quad (2)$$

$$\frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \right] + \rho g_i \quad (3)$$

$$\frac{\partial}{\partial x_j}(\rho C_p u_j T) = \frac{\partial}{\partial x_j} \left(k \frac{\partial T}{\partial x_j} \right) + S_T, \quad (4)$$

where u_i is the velocity, p the pressure, g_i the gravity, and T the temperature. For the material property, ρ is the density, μ the viscosity, c_p the specific heat, and k the thermal conductivity. The density of air is calculated from the ideal gas equation:

$$\rho = \frac{p}{(R/M)T} \quad (5)$$

Here, R is the universal gas constant and M represents the molecular weight of the air. The radiation heat transfer was modeled using the discrete ordinates model (D.O. model):

$$\frac{dI(\vec{r}, \vec{s})}{ds} + (a + \sigma_s)I(\vec{r}, \vec{s}) = an^2 \frac{\sigma T^4}{\pi} + \frac{\sigma_s}{4\pi} \int_0^{4\pi} I(\vec{r}, \vec{s}') \phi(\vec{s} \cdot \vec{s}') d\Omega = 0, \quad (6)$$

where I is the radiation intensity, a is the absorption coefficient, σ_s the scattering coefficient, σ the Stefan-Boltzmann constant, ϕ the phase function, $\vec{r}$ the position vector, $\vec{s}$ the direction vector, and Ω the solid angle in the radiative transfer Eq. (6) [24,25]. Consistent with the authors' previous study on the natural convection [27], the radiation heat transfer rate Q_{rad} was estimated by the equation below [26] and added to the energy source term,

$$Q_{rad} = \frac{\sigma A_t (T_w^4 - T_{surw}^4) \varepsilon}{(1 - \varepsilon) + \frac{A_t}{A_{surw}}}, \quad (7)$$

where A_t is the total surface area of the heat sink, A_{surw} is the surface area of the surrounding air wall around the heat sink, T_w represents the wall temperature of the heat sink, T_{surw} is the temperature of the surrounding air wall, and ε is the emissivity of the heat sink.

The total heat transfer rate and corresponding overall heat transfer coefficient are

$$Q_t = Q_{cond} - Q_{rad} \quad (8)$$

$$\bar{h} = \frac{Q_t}{A(T_w - T_\infty)} \quad (9)$$

The definition of the Grashof number is

$$Gr_L = \frac{g\beta(T_w - T_\infty)L^3}{\nu^2}, \quad (10)$$

where β is the coefficient of thermal expansion and ν is the kinematic viscosity.

In this paper, the length of the blower domain L_{in} is identical to the length of the heat sink L , and the average velocity U_{avg} on the inlet boundary was chosen as the reference velocity. Averaged properties of the fluid were also used in the calculation. With these variables, the Reynolds number is expressed as

$$Re_L = \frac{\rho U_{avg} L}{\mu} = \frac{U_{avg} L}{\nu}. \quad (11)$$

From the assumption of the ideal gas, the coefficient of thermal expansion can be determined from $\beta = 1/T$. In the mixed convective problem, the leading physical parameter is the Richardson number, and it can be expressed using the Grashof and Reynolds numbers:

$$Ri = \frac{Gr}{Re^2} \quad (12)$$

Typically, the range of Richardson number under mixed convection is 0.1–10 [14]. When the Richardson number is less than 0.1, the natural convection can be negligible, while the forced convection is negligible for Richardson numbers larger than 10. In the present study, however, all numerical calculations were conducted under the constant heat flux condition. Therefore, the Grashof number and Richardson number should be modified as shown below.

$$Gr^* = \frac{g\beta q'' L^4}{\nu^2 k} \quad (13)$$

$$Ri^* = \frac{Gr^*}{Re^2} = \frac{g\beta q'' L^2}{\mu^2 k} \quad (14)$$

These are the modified Grashof number and the modified Richardson number, respectively. The range of the modified Richardson number corresponding to the mixed convection regime for the constant heat flux condition is 500–2500 in the present study. The calculation for the pure natural convection case is also included for comparison in this paper.

2.3. Numerical method and validation

Calculations were conducted using the commercial computational fluid dynamics (CFD) software ANSYS Fluent 13 to solve the flow and heat transfer problem. The steady-state calculations were performed with the second-order upwind discretization and the SIMPLE algorithm for velocity-pressure coupling. The Discrete Ordinates (D.O.) method was chosen to calculate the radiation heat transfer. The convergence criterion for the residual of energy equation was set to 10^{-8} , while it was 10^{-6} for the other equations. For radiation, the intensity of discrete ordinate was checked for the convergence. The average temperature on the plate of the heat sink and the mass averaged temperature of air were

also monitored to ensure the convergence of the calculation. The operating fluid is air and corresponding Prandtl number was 0.71 in the numerical simulations.

To validate the accuracy of the numerical approach, the calculated results were compared with the experimental data by Kobus and Oshio. [14]. The result of the mesh independency test for this problem is shown in Fig. 3, representing the variation of the thermal resistance between the heated plate and the air with the number of computational cells. As shown, convergence of the thermal resistance was observed when the computational domain has discretization with 4.8 million cells, and it was considered that the mesh independence condition was enough to ensure the accuracy of the numerical simulations. For the models tested in this paper, the same number of grids per unit spatial length was formed based on this result. As shown in Fig. 4, the numerical results of the thermal resistance with variation of the inlet-air velocity were in good agreement with theoretical predictions and experimental result.

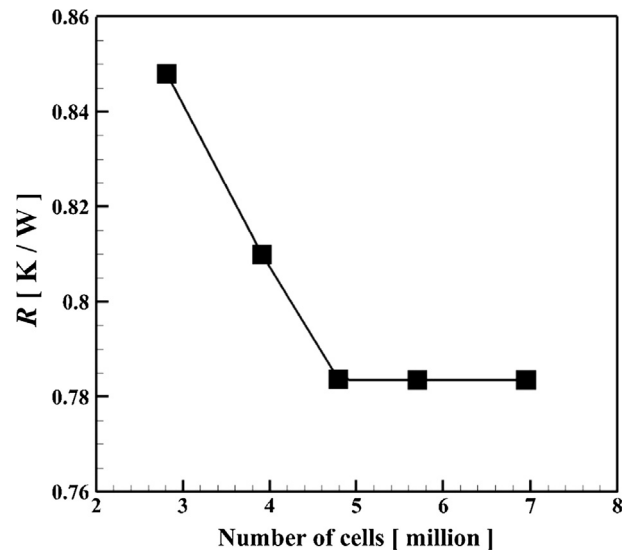


Fig. 3. Result of the mesh independency test for the external mixed convection condition.

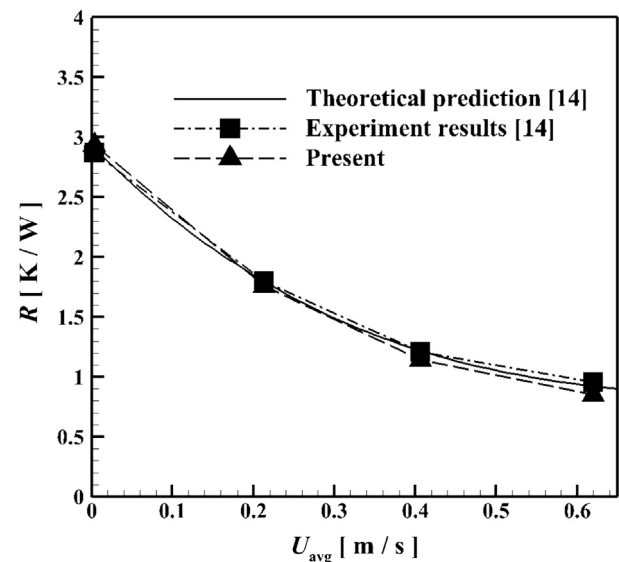


Fig. 4. Comparison of the calculated thermal resistance between the heated plate and air with variation of the inlet velocity.

The difference between the numerical results and the theoretical predictions was less than 7.4%, and the difference between the calculations and experimental data was less than 8.8%.

3. Results and discussion

3.1. Aero-thermal characteristics of slanted-pin fins

Fig. 5 shows the distribution of the normalized temperature on the plane of the fin-side surface with variation of the fin-inclination angle for various modified Richardson numbers. Here, the temperature T^* is normalized by the average wall temperature of the heat sink having horizontal-pin fins ($\theta = 0^\circ$). It is defined as

$$T^* = \frac{T - T_\infty}{\bar{T}_{base} - T_\infty}. \quad (15)$$

In this calculation, the value of the baseline average temperature $\bar{T}_{base}$ is 336.5 K. Overall, the temperature on the fin-side surface tends to decrease as it moves downstream due to the heating flow of the heated plate on the vertical wall. The side surface of the fin in the fin-stem region has a high temperature and decreases along the fin length into the fin tip. As the modified Richardson number decreases, meaning that the forced convection effect increases, the average temperature on the side surface of the fin decreases due to the increasing aiding flow, enhancing the convective heat transfer. In the upstream region, a temperature decrease caused by forced convection is clearly observed for a low modified Richardson number. Moreover, a thermal mixing layer is formed by forced convection at the fin-tip region, and it grows in the downstream direction, as presented in Fig. 5(d). For a large modified Richardson number, this growth of the thermal mixing layer is limited near the leading edge of the cooler region. For the pure natural convection

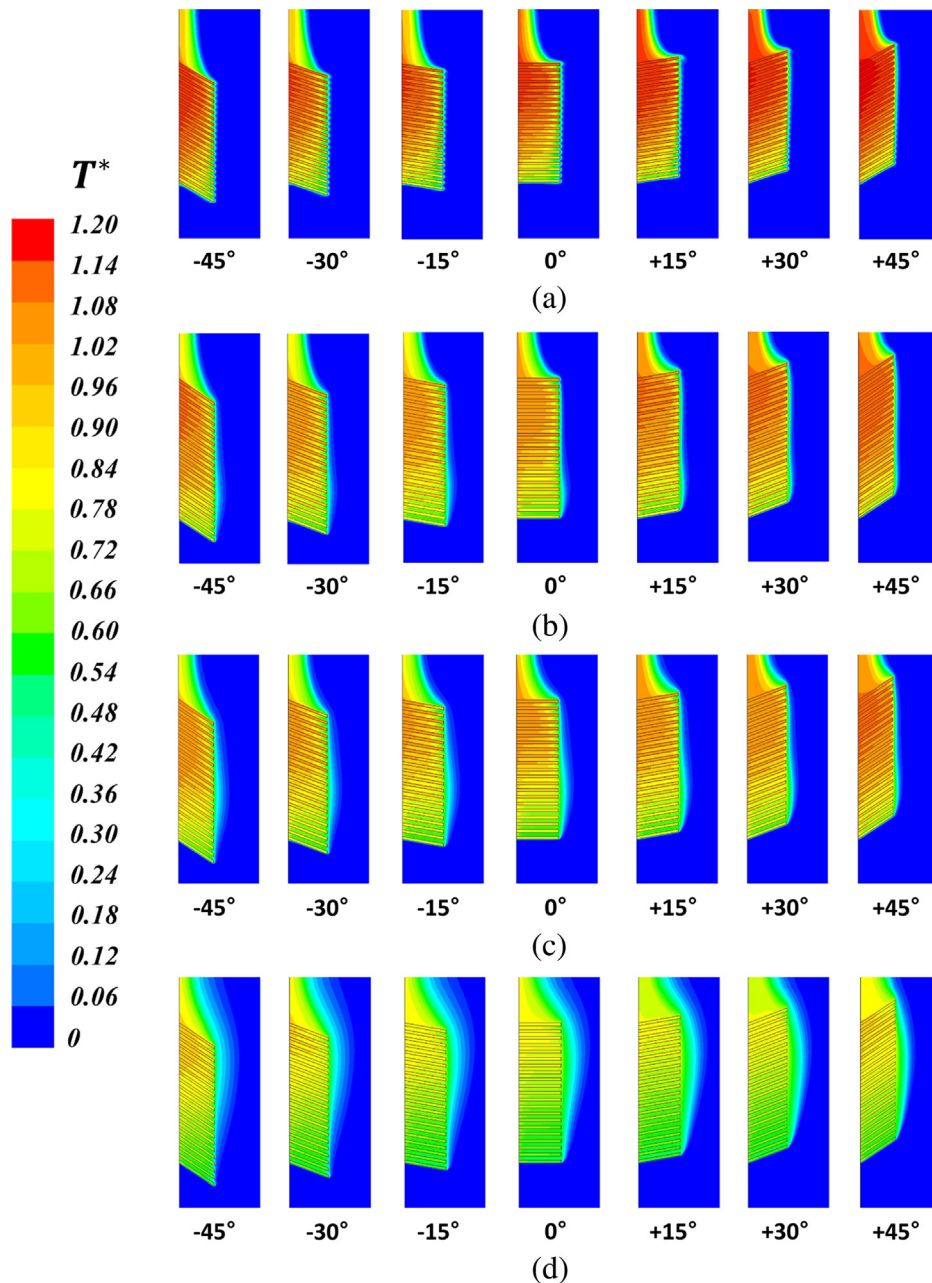


Fig. 5. Distribution of the normalized temperature on the plane of the fin-side surface with variation of the fin-inclination angle for various modified Richardson numbers: (a) $Ri^* = \infty$ (pure natural convection), (b) $Ri^* = 2500$, (c) $Ri^* = 1500$, and (d) $Ri^* = 500$.

case depicted in Fig. 5(a), as the fin-inclination angle increases, the temperature on the side surface of the fin increases. This is the result of the penetration of cold air from the quiescent region for negatively inclined fins, as in the authors' investigation in part-I of this paper [27]. As the modified Richardson number decreases, however, the advantageous fin-inclination angle showing the lowest average temperature changes from the negative inclination angle to the positive one, as shown in Fig. 5(b)–(d). Due to the change of the advantageous inclination angle, it can be expected that there should be a critical Richardson number that alters the sign of the fin-inclination angle for the fin showing the best heat transfer performance.

Fig. 6 is the distribution of the pressure coefficient on the mid-plane between the lateral fins where periodic boundary conditions are imposed for $Ri^* = 1500$ with variation of the fin-inclination angle. Here, the pressure coefficient is defined by

$$C_p = \frac{p}{\frac{1}{2} \rho U_{avg}^2}. \quad (16)$$

For the horizontal fin in Fig. 6(b), the isobaric contour line is nearly parallel to the fin, showing that the direction of the pressure gradient is vertical. For the inclined fin cases, similarly, the direction of the pressure gradient depends on the inclined angle of the fins, and this drives the flow in the negative pressure gradient direction, which is not vertical. For the negatively inclined cases in Fig. 6(a), there is relatively high pressure in the fin-stem region due to the flow resistance at the negative inclined angle. This pressure difference between the fin-stem and fin-tip regions causes the flow to be pushed out into fin-tip region. For positive inclination angles, conversely, the value of the pressure at the fin-stem region becomes small, and it can be predicted that there should be a driving force into the hot plate region, which is similar to the result

with forced convection by Kim et al. [7,8]. This pressure gradient causes the stream to move onto the heated plate, which enhances the heat transfer performance on the base of the cooler.

Given that the driving force of a buoyant flume is the density gradient due to the temperature variation, it is important to observe the temperature distribution, as presented in Fig. 7. In all cases, because the hot plate has the highest temperature, the buoyant flow tends to flow out from the fin stem region to the quiescent cold outer region. This temperature gradient into the fin-tip region is enlarged as the value of the fin-inclination angle changes from negative to positive. This explains how the pushing-out flow from the hot plate to the outside region is enhanced for positively inclined fins. When the forced convection becomes large (a small modified Richardson number), the temperature gradient into the fin-tip region becomes small, and the temperature contours are nearly horizontal in most of the fins, except in the negative-angle cases. As the Richardson number decreases, it affects the temperature distribution and thus weakens the pushing out of the buoyant flume. The flow pattern near the inclined fin should be determined by the relative strengths of the pressure and temperature gradients, as well as by the inertial force of the forced convection effect.

Fig. 8(a)–(c) show the distribution of the pressure coefficient on the fin surface of the models considered here with variation of the modified Richardson number for fin-inclination angles at 45° , 0° , and $+45^\circ$, respectively. In contrast to the conventional forced convection case in Kim et al. [7,8], the pressure does not show a monotonically decreasing pattern along the stream. The pressure, instead, increases from the leading edge of the cooler and decreases after the peak value in the middle part of the cooler. The peak value of the pressure occurs at the fin-stem region for the negative fin-inclination angle and at the fin-tip region for the positive fin-inclination angle. This pressure pattern is similar to

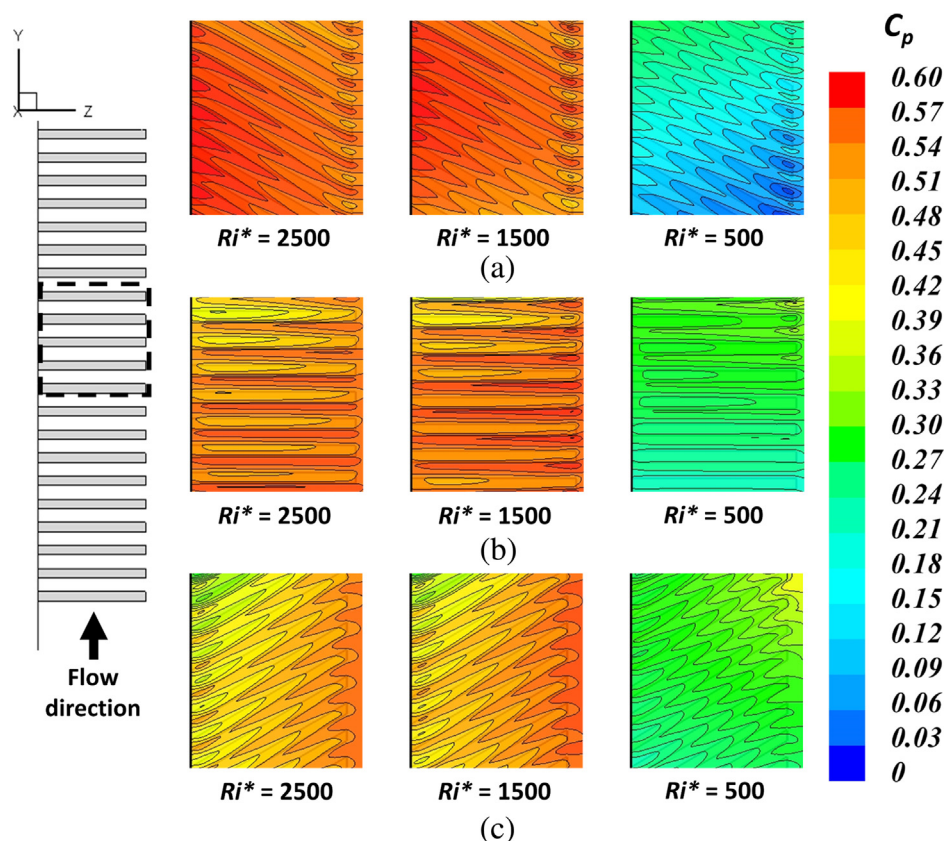


Fig. 6. Distribution of the pressure coefficient on the mid-plane between lateral fins for $Ri^* = 2500$, 1500 and 500 with variation of the fin-inclination angle: (a) $\theta = -45^\circ$, (b) $\theta = 0^\circ$, and (c) $\theta = +45^\circ$.

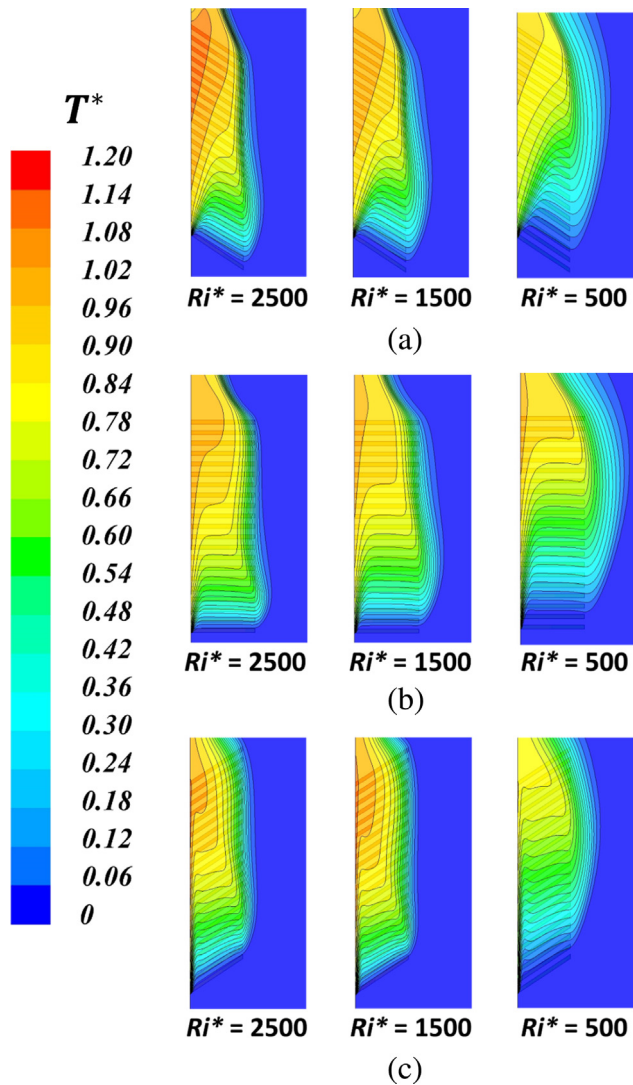


Fig. 7. Distribution of the normalized temperature on the mid-plane between lateral fins for $Ri^* = 2500$, 1500 and 500 with variation of the fin-inclination angle: (a) $\theta = -45^\circ$, (b) $\theta = 0^\circ$, and (c) $\theta = +45^\circ$.

the results in Fig. 6. As the pressure gradient in the vertical direction changes its sign from adverse to favorable, the flow decelerates from the leading edge of the cooler but accelerates after the peak point. It can also be predicted that the flow to/from the heated plate can change after the peak-value location.

Fig. 9 shows the distribution of the skin-friction factor on the fin surface. Here, the skin friction factor is defined as

$$C_f = \frac{\tau_w}{\frac{1}{2} \rho U_{avg}^2}, \quad (17)$$

where τ_w is the wall-shear stress on the surface of the fin. The wall shear stress shows the opposite trend compared to the result of the pressure variation shown earlier in Fig. 8. Along the vertical direction, the skin-friction factor decreases from the leading edge of the cooler and increases again after the local minimum value. In the horizontal direction, the maximum value is noted at the fin-tip region for the negative angle and at the fin-stem region for the positive angle. This clearly indicates that the flow acceleration into the heated plate for a positively inclined fin produces high skin friction, especially near the fin at the trailing edge of the cooler. The negatively inclined fin, however, has high skin friction in the fin-tip region.

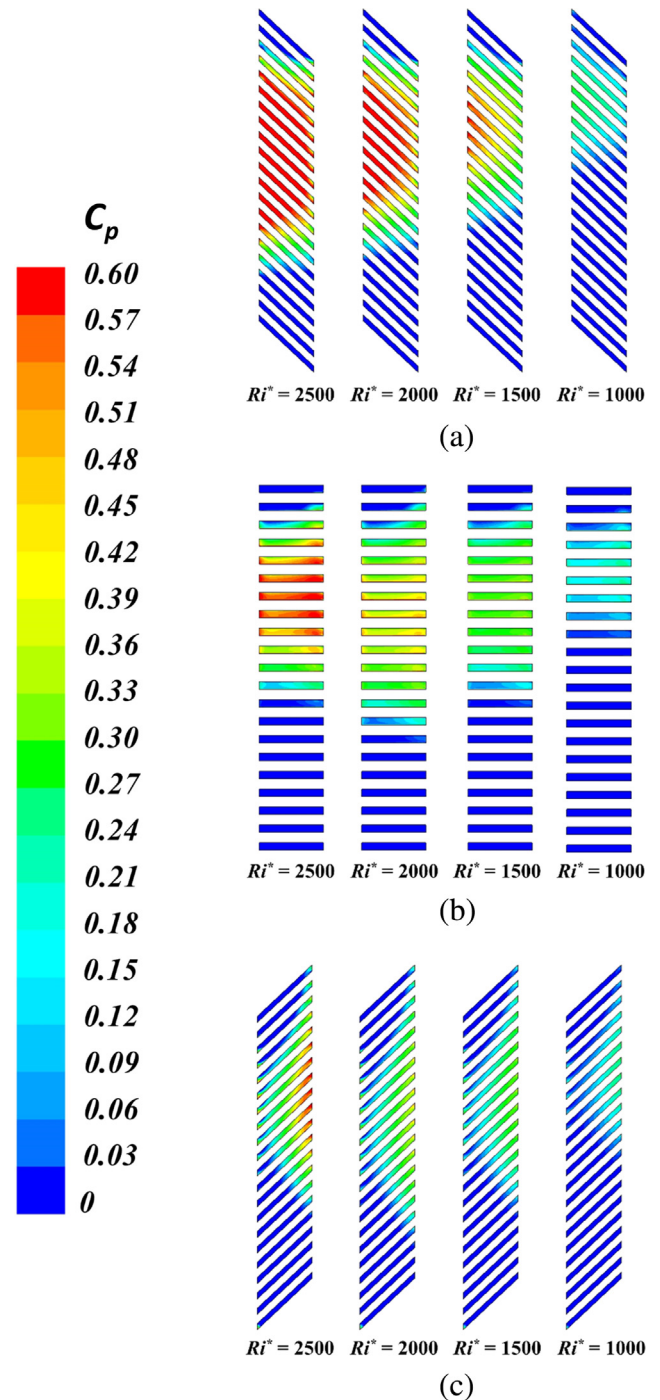


Fig. 8. Distribution of the pressure coefficient on the fin surface with variation of the modified Richardson number for: (a) negatively inclined fins ($\theta = -45^\circ$), (b) horizontal fins ($\theta = +0^\circ$), and (c) positively inclined fins ($\theta = +45^\circ$).

3.2. Flow patterns around the fins

Fig. 10 shows the flow path-lines colored according to the fluid temperature on the mid-plane between the lateral fins. As predicted from the results of the pressure and temperature gradients shown in Figs. 6–8, the pushing-out flow into the fin-tip region for the negative fin-inclination angle and the acceleration of the flow into the fin-stem direction, as well as its local variation of the trend switching from deceleration to acceleration around the peak values of the pressure, are clearly captured. As the forced convection effect increases, for a small modified Richardson number, the cur-

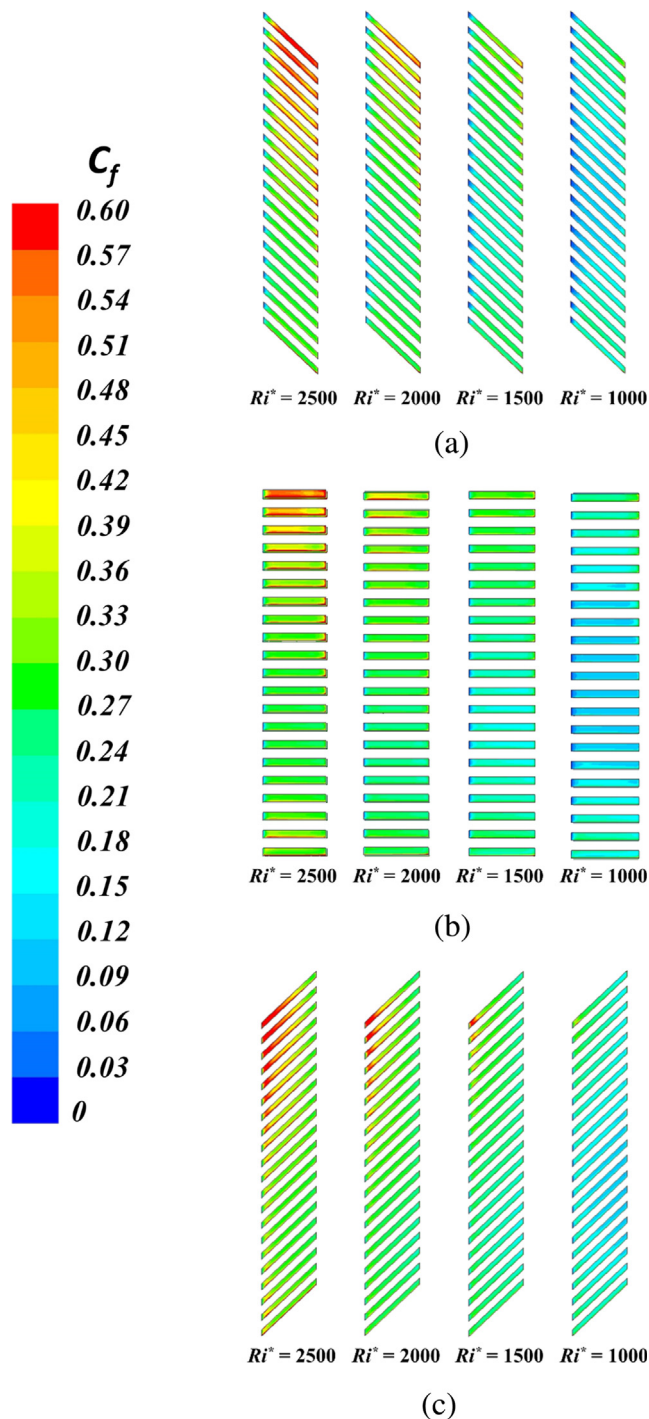


Fig. 9. Distribution of the skin-friction factor on the fin surface with variation of the modified Richardson number for: (a) a negatively fin-inclined angle ($\theta = -45^\circ$), (b) horizontal fins ($\theta = +0^\circ$), and (c) a positive fin-inclination angle ($\theta = +45^\circ$).

vature of the path-line becomes large, as the inertial force and corresponding pressure gradient are relatively strong. For the negative inclination angle shown in Fig. 10(a), the result of the $Ri^* = 2500$ case shows a more attached flow into the heated plate, whereas the flow pattern for the $Ri^* = 500$ case is more attached than other cases for the positive fin-inclination angle. The peak point where the flow direction turns into the heated plate also varies; it goes to the downstream region of the cooler for the negative angle and to the upstream region for the positive angle as the modified Richardson number decreases. This location does not precisely

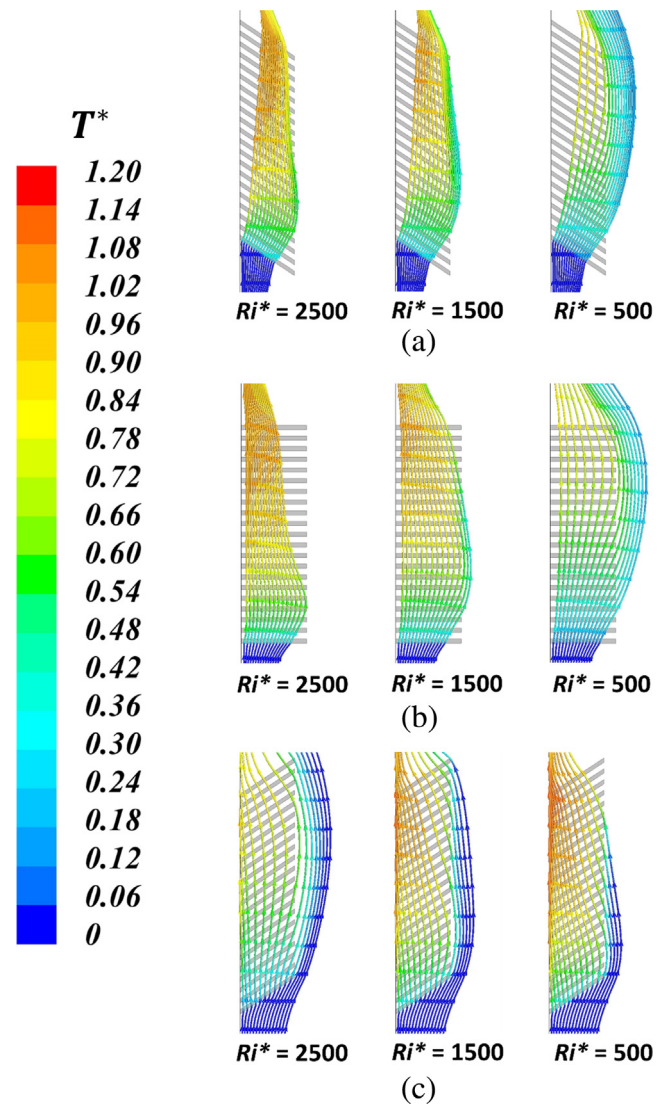


Fig. 10. The flow path-lines colored by the fluid temperature on the mid-plane between the lateral fins for various modified Richardson numbers with variation of the fin-inclination angle: (a) negative fin-inclination angle ($\theta = -45^\circ$), (b) zero fin-inclination angle ($\theta = +0^\circ$), and (c) positive fin-inclination angle ($\theta = +45^\circ$).

coincide with the peak point location of the pressure due to the incoming inertial force effect of the flow. Based on these transitions of the flow pattern depending on the forced convection effect (Richardson number) and the fin-inclination angle, the heat transfer performance of the cooler should be varied.

In the authors' previous work on the natural convection study for slanted-pin fins [27], it was found that the flow in the vertical fin interspacing and corresponding heat transfer coefficient on the upper and lower surfaces of a rectangular fin has a considerable effect on the total heat transfer performance. In Fig. 11, several fluid particle trajectories that start from the inlet of the computational domain are plotted for $Ri^* = 2500$, 1500, and 500. As shown in the figures, the trajectories show complex flow patterns, especially along the narrow fin-interspacing region. For the negative fin-inclination angle ($\theta = -45^\circ$) in Fig. 11(a), a penetrating flow in the fin interspacing into the fin-stem region is observed. According to the authors' previous study on the natural convection, this penetration of a cold flow from the outer quiescent region into the heated plate enhances the heat transfer coefficient in the region. In the lateral gap of the fins, however, only the rising flow into the fin-tip region is shown, which is similar to the result in

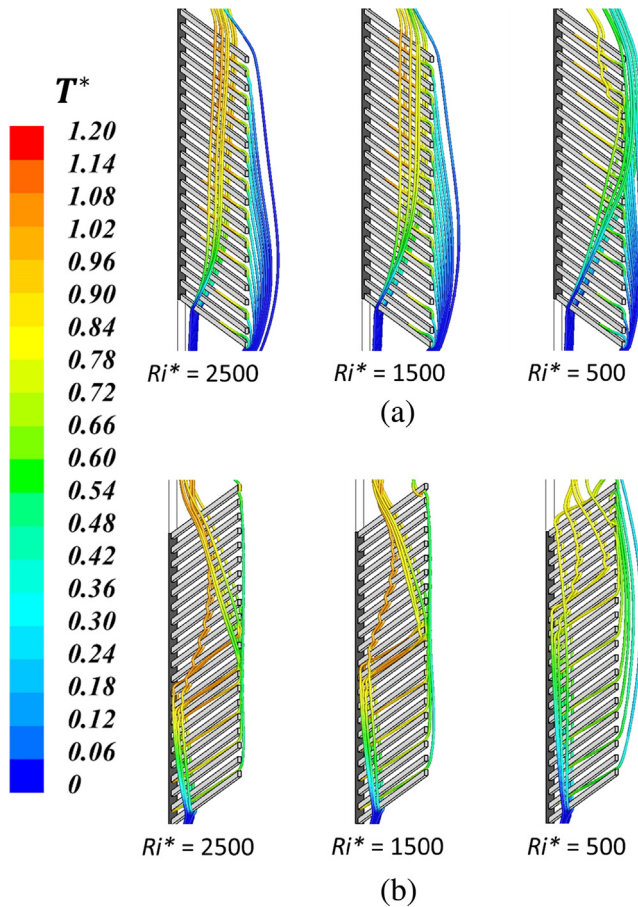


Fig. 11. The trajectories that individual fluid particles follow from the inlet: (a) negative fin-inclination angle ($\theta = -45^\circ$), and (b) positive fin-inclination angle ($\theta = +45^\circ$).

Fig. 10. For the positive fin-inclination angle ($\theta = +45^\circ$) shown in Fig. 11(b), the flow pattern is reversed, and the rising hot flow is captured in the vertical fin-interspacing region. For the flow along the lateral gap of the fins, the inflow impinges upon the heated plate at some distance from the leading edge of the cooler and moves upward to the fin-tip region, aiding the pushing-out hot flow between the vertical fin interspacing. The length of the impingement on the heated plate becomes longer as the modified Richardson number decreases due to the increasing inertial force of forced convection. As a result, the heat transfer coefficient on the heated plate should be increased for a small Richardson number.

3.3. Overall heat transfer performance

The distribution of the normalized heat transfer coefficient on the side surface of fins is plotted in Fig. 12. Similar to the normalized temperature in Fig. 5, the local heat transfer coefficient is normalized by the horizontal pin-fin model for comparison.

$$h^* = \frac{h}{h_{base}} \quad (18)$$

The value of the baseline average heat transfer coefficient $\bar{h}_{base}$ in this calculation is $5.09 \text{ W/m}^2 \text{ K}$. For the negatively inclined fins in Fig. 12(a), the surfaces of the fins in the upstream region have a high local heat transfer coefficient because the cold inflow impinges on the fin at the leading edge of the fin, gradually decreasing as the mixing layer between the fin and the outer region develops. As shown in Fig. 10(a), most of the flow from the inlet converges to

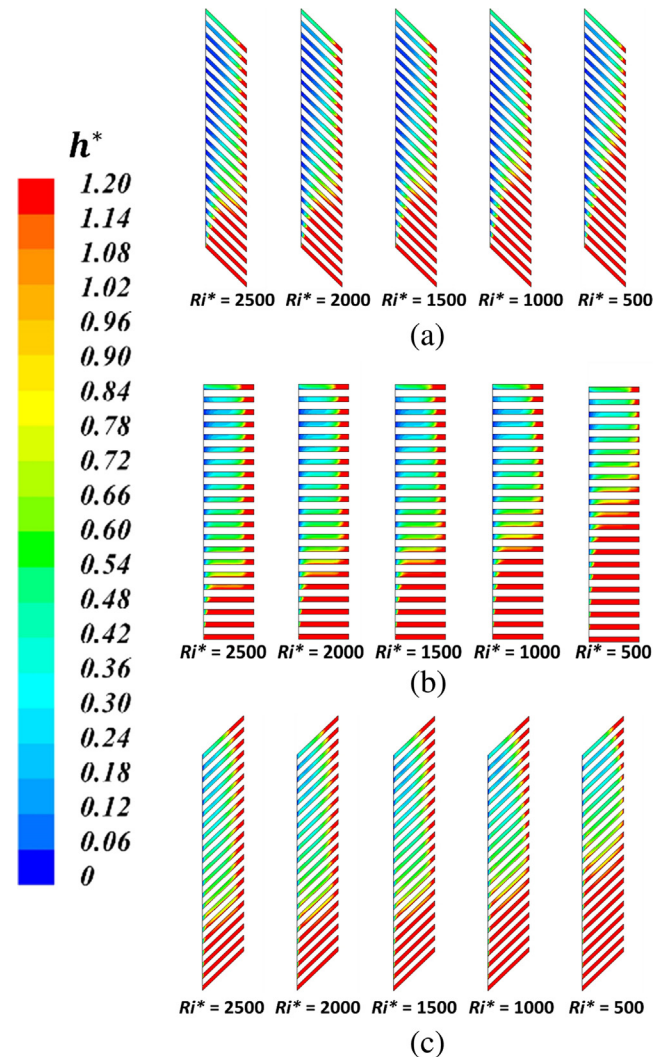


Fig. 12. Distribution of the normalized heat transfer coefficient on the surfaces of the fins for: (a) negatively inclined fins ($\theta = -45^\circ$), (b) horizontal fins, and (c) positively inclined fins ($\theta = +45^\circ$).

the fin-tip region along the fins, and this results in a large value of h^* at the tips of the fins. The side surface in the fin-stem region has a low local heat transfer coefficient due to the influence of hot flow flowing to the fin-tip region. As the modified Richardson number increases, the average heat transfer coefficient increases due to the influence of the forced convection that supplies more cooling air for heat transfer. The horizontal pin fin (Fig. 12(b)), and the negatively inclined pin fin (Fig. 12(c)) have similar distributions of the heat transfer coefficient, showing high heat transfer coefficients on the fins in the upstream region and at the tips of the fins. Furthermore, the heat transfer coefficient on the fin-tip surface increases again for the positive fin-inclination angles in the downstream direction, stemming from the result of the accelerating flow shown in Fig. 10(c). However, in the fin-stem region, the relatively large heat transfer coefficients compared to those in Fig. 10(a) are distributed on the surface owing to the effect of flow forwarding to the base due to the downward pressure gradient component into the heated plate.

Consequently, the heat transfer performance on the base plate shows differences in Fig. 13. Heat sinks with negatively inclined pin fins have poor heat transfer coefficients on the base plate, even with an increase in the Richardson number. As the fin-inclination angle increases, however, the heat transfer performance of the base

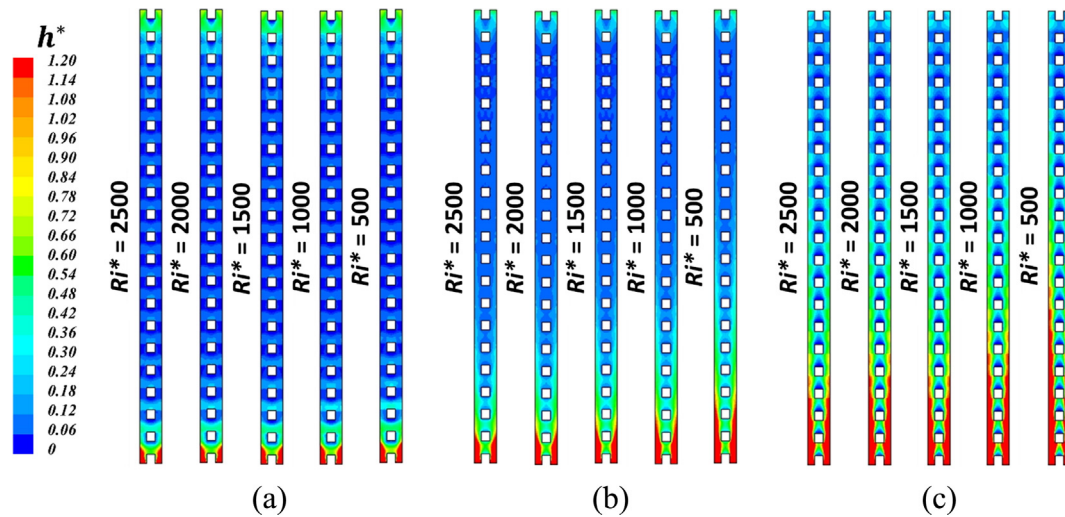


Fig. 13. Distribution of the normalized heat transfer coefficient on the surface of the base plate of the cooler for: (a) negatively inclined fins ($\theta = -45^\circ$), (b) horizontal fins, and (c) positively inclined fins ($\theta = +45^\circ$).

plate is clearly improved as the modified Richardson number decreases.

The average heat transfer coefficient on the surface of the vertical fin interspacing is plotted in Fig. 14. As in our previous study on the natural convection [27], this surface is affected by the buoyant flow, as shown in Fig. 11. Due to the cold penetrating flow for the negative fin-inclination angle, the heat transfer coefficients in Fig. 14 show large values compared to the positively inclined fin results. As the Richardson number decreases, however, the advantage in the heat transfer coefficient in this region becomes smaller, whereas the heat transfer coefficients increase for a positive fin-inclination angle. Although the resulting heat transfer coefficient of a negative angle for the smallest Richardson number of $Ri^* = 500$ still has a larger value than that of the positive angle, the advantage of this region with regard to the total heat transfer rate becomes small for the large forced convection case.

Fig. 15 shows the variations of the normalized average heat transfer coefficient of a slanted pin-fin heat sink with various

modified Richardson numbers. In this figure, to evaluate the contribution of each part to the heat transfer performance, the corresponding values at the fin surface, the bottom heated plate and the total heat sink surface are plotted separately. On the fin surface in Fig. 15(a), the negatively inclined fins show better performance, as it close to the natural convection for large modified Richardson numbers, due to the cold flow penetration effect. For $Ri^* = 500$, however, the horizontal fin with $\theta = 0^\circ$ shows the largest heat transfer coefficient on the fin surface. On the bottom heated plate in Fig. 15(b), as shown in Fig. 13, the variation of the heat transfer coefficient is very sensitive to the fin-inclination angle, and it increases as this angle increases due to the accelerated flow near the fin-stem region. The total averaged heat transfer coefficient for the entire heat sink is plotted in Fig. 15(c). In general, the heat transfer performance improves as the forced convection effect increases. Furthermore, as expected, the transition of the fin-inclination angle showing the best heat transfer performance changes as the modified Richardson number decreases. Finally, to

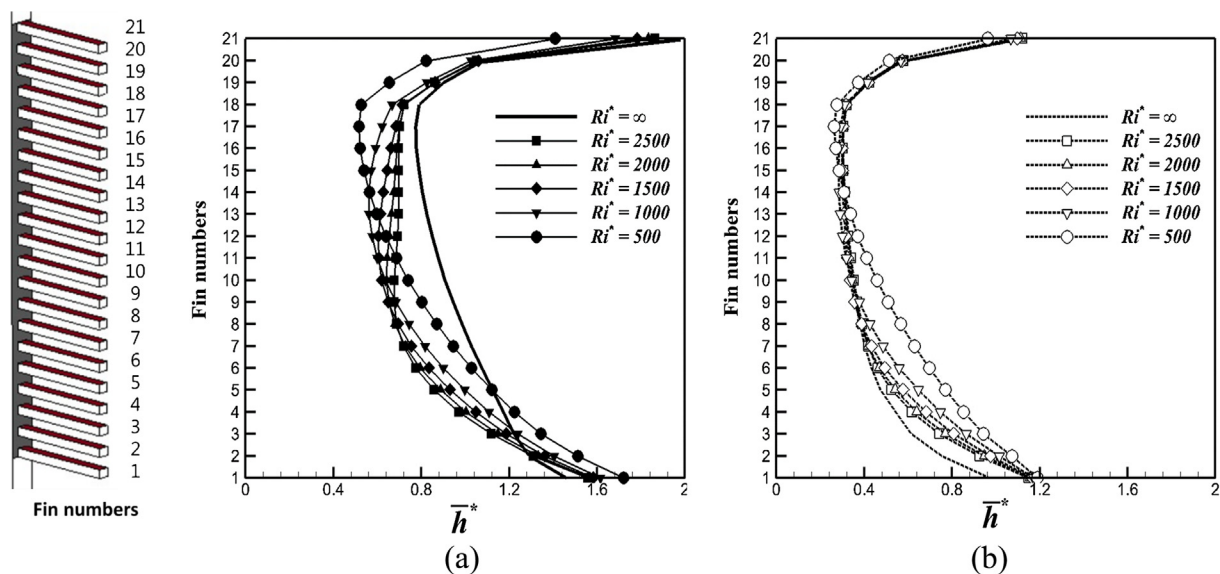


Fig. 14. The average heat transfer coefficient on the surface of the vertical fin interspacing: (a) negative fin-inclination angle ($\theta = -45^\circ$), and (b) positive fin-inclination angle ($\theta = +45^\circ$).

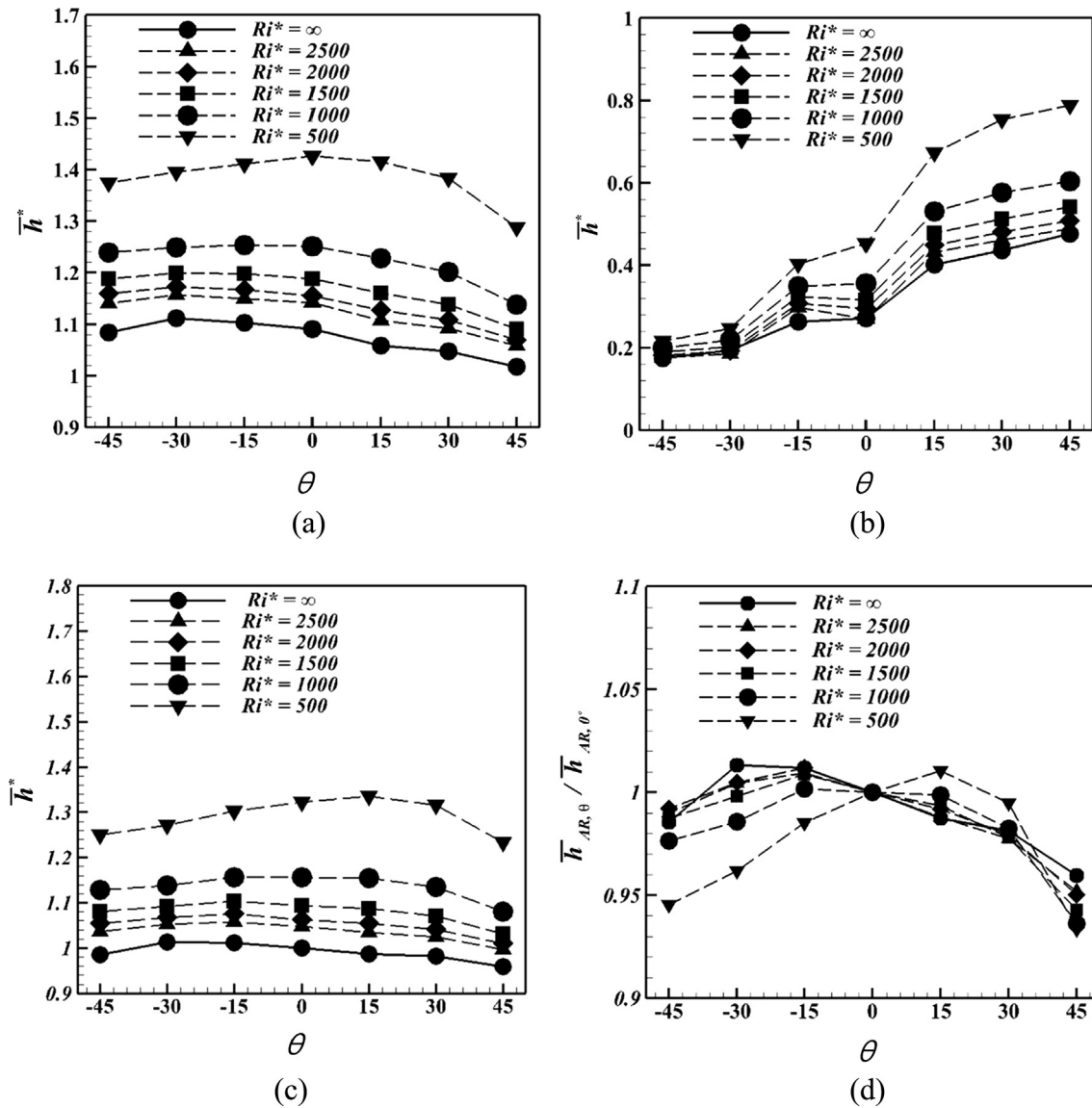


Fig. 15. Comparison of the average normalized heat transfer coefficients of the slanted pin-fin heat sink for various modified Richardson numbers at: (a) the fin surface, (b) the bottom heated plate, (c) the total surface of the entire heat sink, and (d) the normalized heat transfer coefficient of the heat sink.

observe this transition more clearly, the heat transfer coefficients are normalized by the value of the horizontal fin shown in Fig. 15(d). Among the fins tested in this paper, the best performance is captured with fins for which $\theta = -30^\circ$ for the natural convection condition. At $Ri^* = 500$, however, the fins having $\theta = +15^\circ$ shows the best performance. The critical modified Richardson number with the horizontal fin ($\theta = 0^\circ$) showing the best performance was close to $Ri^* = 1000$. This transition is based on the effect of the weaker convection effect in the vertical fin interspace region for the negative fin-inclination angle and on the contribution of the bottom heated plate to the total heat transfer rate caused by the accelerated flow near the fin-stem region for the positive fin-inclination angle.

4. Conclusion

The heat transfer performance capabilities of slanted-pin fins having inclination angles in the stream-wise direction were studied numerically under a mixed convection condition in the laminar

flow regime. In addition to the authors' accompanying paper [27] on the natural convection, the mixed convection condition was considered by varying the modified Richardson number for the range of 500–2500. The three-dimensional flow and heat transfer problem was solved using a simplified computational model utilizing periodic boundary conditions in the fin lateral direction.

As in the natural convection case, for a large Richardson number, the penetration of cold air into the vertical fin interfacing was captured. Moreover, an accelerated flow into the fin-stem region was observed for positively inclined fins. It was found that the relative strengths of these flows are determined by three main driving forces: (i) the pressure gradient along the pin fins, (ii) the temperature gradient and corresponding buoyant flow, and (iii) the inertial force of the incoming forced convection component. For negative fin-inclination angles, the flow is pushed out to the fin-tip region due to the high pressure near the bottom plate. For positive fin-inclination angles, however, the flow goes to the heated plate owing to the secondary pressure gradient to the fin-stem region. The pressure gradient shows adverse and favorable patterns in the vertical direction, resulting in decelerating and

accelerating flows along the vertical direction. The change of the flow pattern depends on the modified Richardson number; as the forced convection becomes stronger, the point of acceleration moves in the downstream direction. In the vertical fin-interspacing region, cold air penetration from the quiescent region was captured for the negative fin-inclination angle, whereas, hot air flowing out was observed for the positive fin-inclination angle. As the modified Richardson number changes, the relative importance of these flows varies.

The positive inclination angle has advantage in the heat transfer under the forced-convection condition due to the flow acceleration into the heated-plate region, as in Kim et al. [7,8]. For the natural convection condition, however, the inclination angle showing the best performance was the negative angle because of the penetrating cold-air flow to the vertical fin-interspacing region, as in the Part-I of this paper [27]. In this paper, for the mixed-convection condition, it was found that the angle of the best performance varies between the negative and positive values depending on the Richardson number. The transition of the best fin-inclination angle for the heat transfer performance has been clearly identified, and the critical modified Richardson number showing the horizontal fin ($\theta = 0^\circ$) for the best performance was close to $Ri^* = 1000$ among the pin-fin models tested in this paper.

The results in this paper and in the authors' previous studies summarize the heat transfer performance capabilities of rectangular slanted pin fins for natural and mixed convection conditions under laminar flow regime. Additional studies on the turbulent flow condition as well as the effects of other geometric parameters, such as the staggered fin arrangement and various fin spacings, should be conducted to build on the findings here.

Conflict of interest

The authors declared that there is no conflict of interest.

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